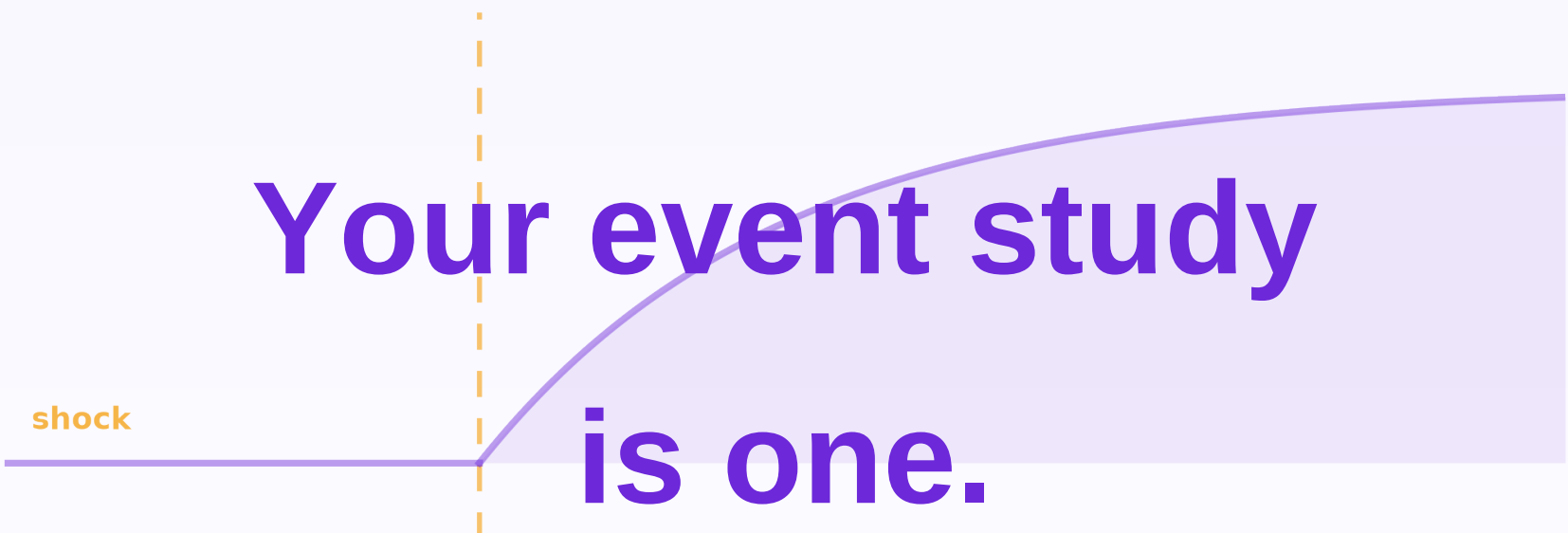


diff - diff

**Macro built local projections to read
the dynamic effects of shocks.**



**Your event study
is one.**

LP-DiD: Local Projections meet DiD. Now in diff-diff.

Dube, Girardi, Jordà & Taylor (2025), J. of Applied Econometrics 40(5).

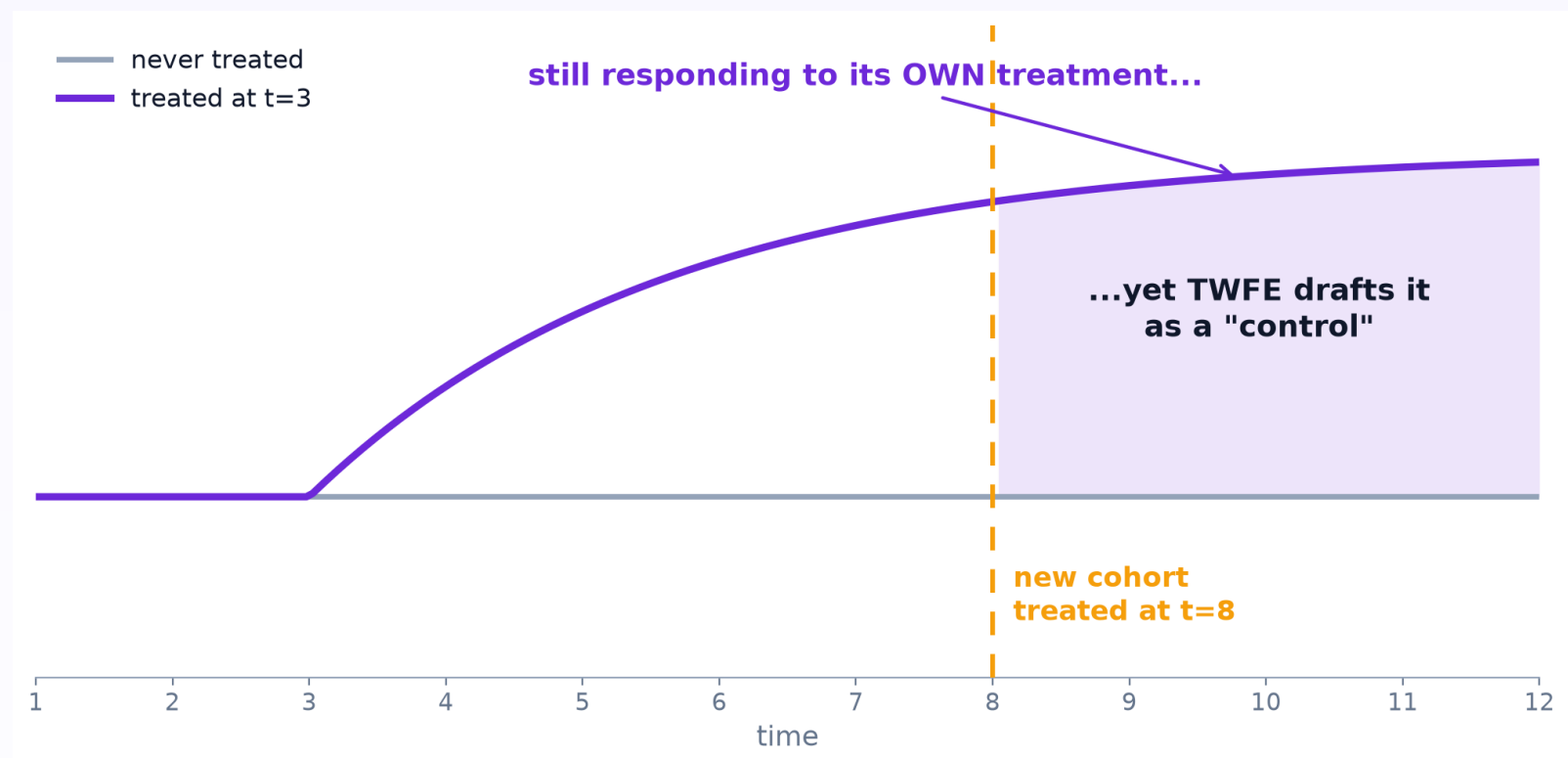
Co-author Òscar Jordà introduced local projections in 2005.

THE PROBLEM

Already-treated units sneak into your control group.

Previously treated units, which might still be experiencing lagged ... treatment effects, are implicitly used as controls for newly treated ones.

- Dube, Girardi, Jordà & Taylor (2025)



The TWFE estimate could even lie outside the range
of group-specific treatment effects.

So just point macro's tool
at your rollout?

Not so fast.

A naive LP implementation inherits the same
negative-weighting bias as TWFE.

(Dube, Girardi, Jordà & Taylor 2025, Eqs. 6-7)

The estimator isn't the fix.

The comparisons are.

The underappreciated feature.



An underappreciated feature of the LP framework is that it is straightforward to limit the set of permissible comparisons based on a desired criterion, such as past treatment history.

- Dube, Girardi, Jordà & Taylor (2025)

Local projections

Jordà (2005). Built to estimate
dynamic average effects.



Clean controls

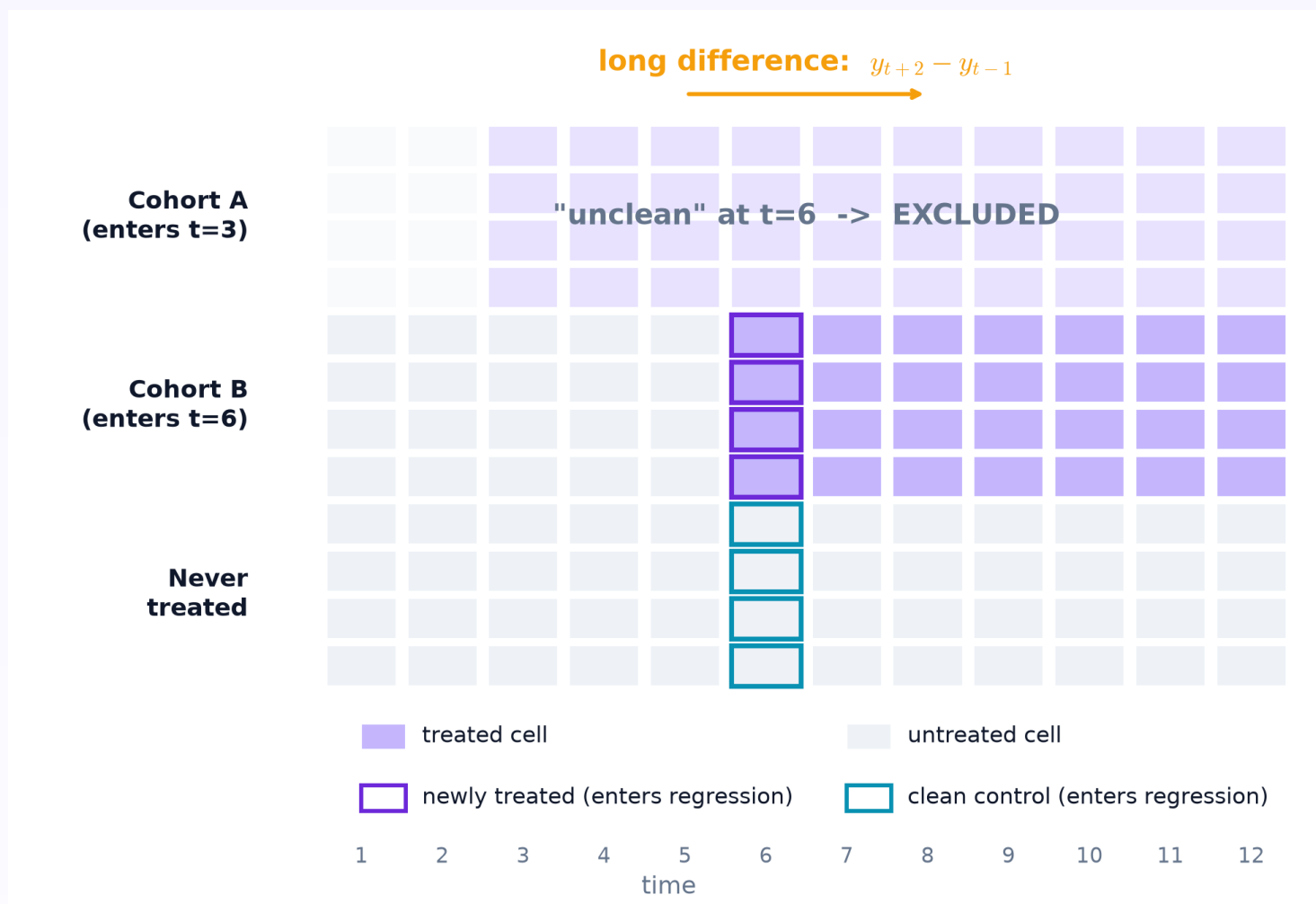
Cengiz et al. (2019) spirit.
"Unclean" units stay out.



LP-DiD

Dynamic treatment effects with provably non-negative weights.

"Unclean" observations leave the control group.



Each horizon is one regression you could run by hand.

In the absorbing (default) path, already-treated units never serve as controls.

Non-negative weights, guaranteed.

$$y_{i,t+h} - y_{i,t-1} = \beta_h \Delta D_{it} + \delta_t^h + e_{it}^h$$

$$E(\beta_h) = \sum_{g \neq 0} \omega_{g,h} \tau_h^g, \quad \omega_{g,h} \geq 0$$

always non-negative -- no forbidden comparisons

And it "subsumes many of the recent solutions" - the abstract, verbatim:

reweight=True

== Callaway-Sant'Anna

exact (Sec. 3.7)

default (variance-weighted)

== Cengiz et al. stacked

exact (Sec. 3.7)

h=0, single cohort

== classic 2x2 DiD

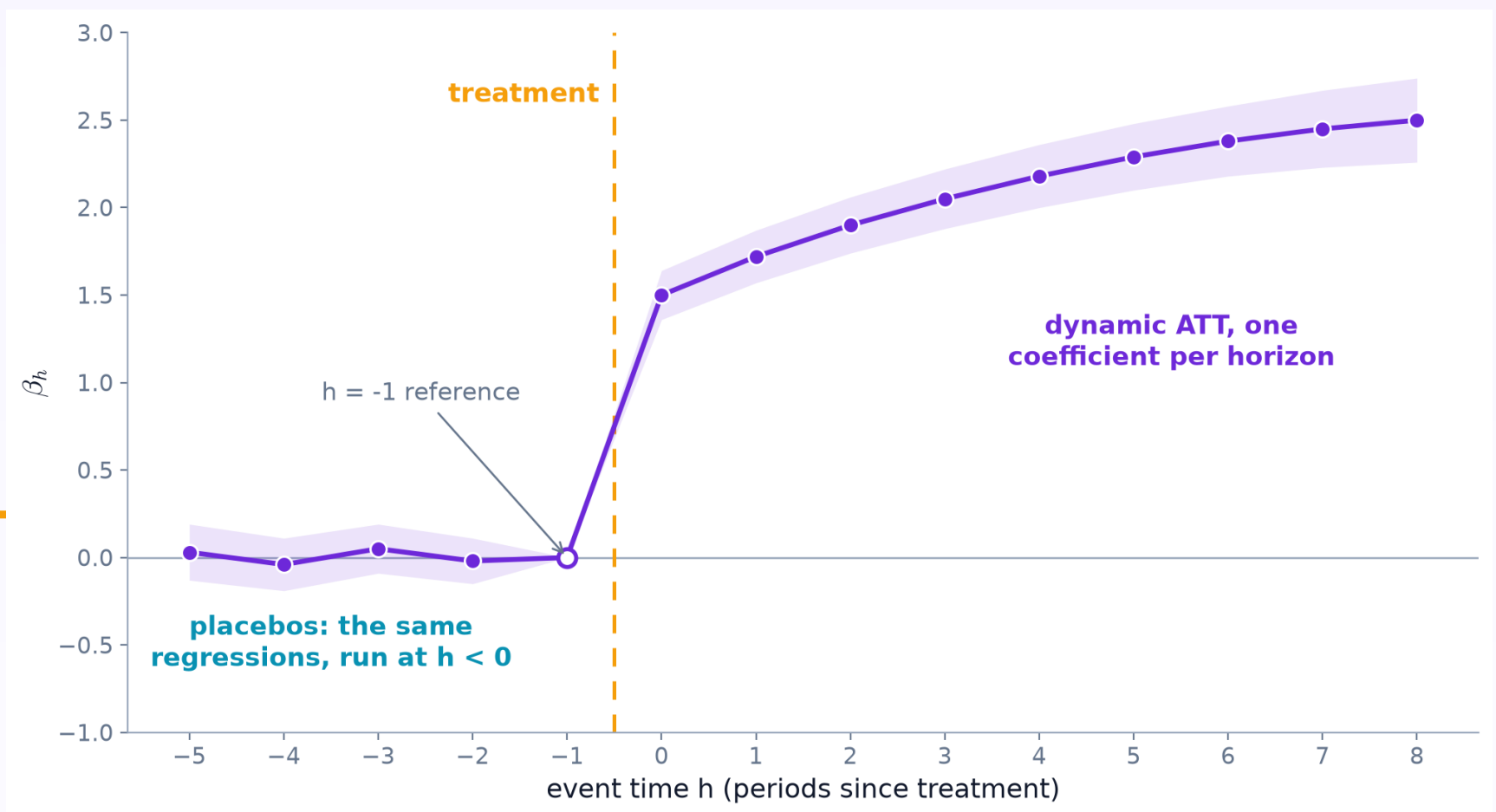
exact (Sec. 2.2)

pmd='max'

-> BJS imputation

exact single-cohort (fn. 10-11)

Pre-trends, estimated symmetrically.



The event study is native - not a post-hoc aggregation.

Pre and post coefficients are estimated symmetrically - the paper notes this

avoids the pretrend-interpretation difficulties raised in Roth (2024).

It's just a regression.

Same sklearn-like API as every diff-diff estimator.

```
from diff_diff import LPDiD

result = LPDiD(
    pre_window=5, post_window=10,
).fit(
    data,
    outcome='revenue', unit='store',
    time='week', treatment='treated',
)

print(result.att)          # pooled post ATT = 2.31
result.event_study         # one beta per horizon

# flip one switch:
#   reweight=True    -> Callaway-Sant'Anna
#   non_absorbing=   -> on/off treatments (entry effects)
#   survey_design=   -> pweights + strata + PSUs (default path)
```

Callaway-Sant'Anna is literally a keyword argument.

IN PRODUCTION

Production-ready.

Cluster-Robust SEs

Unit-level by default,
matches Stata lpdid

Covariates, Two Ways

RA path (BJS-style) or
direct with guardrails

PMD Baselines

Premean differencing +
pooled pre/post ATTs

Non-Absorbing Treatment

First-entry & effect-
stabilization (Eq. 12/13)

Survey Designs

pweights, strata, PSU, FPC;
Binder TSL (default path)

Composition Control

no_composition fixes the
post-window sample

Composable where validated - survey_design runs the default path only.

— VALIDATED —

Validated.

Against the authors' own tooling - and against our own shelf.

Authors' R recipes (danielegirardi/lpdiid)

Event-study and pooled estimands match to $\sim 1e-12$.

Independent fixest reconstruction

Non-absorbing Eq. 12/13, variance-weighted: point + SE to $\sim 1e-13$.

survey::svyglm, end to end

Default-path survey: per-horizon point, SE, and df all pinned.

Equivalences tested, not cited

reweight == our CallawaySantAnna; PMD == our ImputationDiD (1 cohort).

Now in diff-diff.

LP-DiD.

```
$ pip install --upgrade diff-diff
```

github.com/igerber/diff-diff

diff - diff

Difference-in-Differences for Python